A Proof of Theoretical Results

A.1 Proof of Lemma 1

Proof. We first derive the optimal solution for cross-view denoising from the VLA view to the SLA view. To this end, let's consider the mean squared denoising loss for all possible training samples, formulated as

$$\mathcal{L}_{D,v\to s} = \mathbb{E}\|Z_s - h(Z_v)\|^2, \tag{22}$$

where for any function $h(\cdot)$, we can decompose the residual as

$$Z_s - h(Z_v) = \big(Z_s - \mathbb{E}[Z_s \mid Z_v]\big) + \big(\mathbb{E}[Z_s \mid Z_v] - h(Z_v)\big). \tag{23}$$

Substituting Eq. (23) back to Eq. (22), we obtain

$$\begin{aligned}\mathcal{L}_{D,v\to s} = &\ \mathbb{E}\|Z_s - \mathbb{E}[Z_s \mid Z_v]\|^2 + \mathbb{E}\|\mathbb{E}[Z_s \mid Z_v] - h(Z_v)\|^2 \\ &+ 2\,\mathbb{E}\big[\langle A, B\rangle\big],\end{aligned} \tag{24}$$

where $A = Z_s - \mathbb{E}[Z_s \mid Z_v]$, $B = \mathbb{E}[Z_s \mid Z_v] - h(Z_v)$, and $\langle\cdot,\cdot\rangle$ represents the inner product.

Since B is a function of Z_v, based on the basic properties of conditional expectation, we have

$$\mathbb{E}\big[\langle A, B\rangle\big] = \mathbb{E}\Big[\mathbb{E}\big[\langle A, B\rangle \mid B\big]\Big] = \mathbb{E}\Big[\mathbb{E}\big[\langle A, B\rangle \mid Z_v\big]\Big], \tag{25}$$

where

$$\mathbb{E}\big[\langle A, B\rangle \mid Z_v\big] = \langle \mathbb{E}[A \mid Z_v], B\rangle. \tag{26}$$

Note that

$$\mathbb{E}[A \mid Z_v] = \mathbb{E}\Big[Z_s - \mathbb{E}[Z_s \mid Z_v] \mid Z_v\Big] = \mathbb{E}[Z_s \mid Z_v] - \mathbb{E}[Z_s \mid Z_v] = 0. \tag{27}$$

Substituting Eq. (27) back to Eq. (26) and Eq. (25), we obtain

$$\mathbb{E}\big[\langle A,B\rangle\big] = 0. \tag{28}$$

Thus, Eq. (24) is equivalent to

$$\mathcal{L}_{D,v\to s} = \mathbb{E}\|Z_s - \mathbb{E}[Z_s \mid Z_v]\|^2 + \mathbb{E}\|\mathbb{E}[Z_s \mid Z_v] - h(Z_v)\|^2. \tag{29}$$

Since the second term in the right-hand side of Eq. (29) is non-negative, the loss is minimized if and only if

$$h(Z_v) = \mathbb{E}[Z_s \mid Z_v], \tag{30}$$

which corresponds to the optimal denoised SLA view as

$$\widetilde{Z}_s^* = \mathbb{E}[Z_s \mid Z_v]. \tag{31}$$

Similarly, the optimal denoised view in the symmetric direction from SLA to VLA yields

$$\widetilde{Z}_v^* = \mathbb{E}[Z_v \mid Z_s]. \tag{32}$$

□

A.2 Proof of Proposition 1

Proof. Under Assumption 1, the optimal denoised SLA view, i.e., $\widetilde{Z}_s^* = \mathbb{E}[Z_s \mid Z_v]$ shown in Lemma 1, can be rewritten as

$$\widetilde{Z}_s^* = \mathbb{E}[Z_s \mid Z_v] = \mathbb{E}\big[\mathbb{E}[Z_s \mid C, Z_v] \mid Z_v\big], \tag{33}$$

where the expectation $\mathbb{E}[Z_s \mid C, Z_v] = \mathbb{E}[Z_s \mid C]$ due to the conditional independence as $Z_s \perp Z_v \mid C$. Let $f_s(C) \triangleq \mathbb{E}[Z_s \mid C]$, we obtain

$$\widetilde{Z}_s^* = \mathbb{E}[f_s(C) \mid Z_v]. \tag{34}$$

Similarly, let $f_v(C) \triangleq \mathbb{E}[Z_v \mid C]$, the optimal denoised VLA view satisfies

$$\widetilde{Z}_v^* = \mathbb{E}[f_v(C) \mid Z_s]. \tag{35}$$

□

A.3 Proof of Theorem 1

Without loss of generality, we consider the SLA view $(Z_s, \widetilde{Z}_s)$, while the prove is similar for the VLA view $(Z_v, \widetilde{Z}_v)$.

Following Lemma 1 and under Assumption 1, we have

$$Z_s \perp Z_v \mid C. \tag{36}$$

Note that $\widetilde{Z}_s = h(Z_v)$ is a deterministic function of Z_v. Thus, we can further derive that

$$\widetilde{Z}_s \perp Z_s \mid C \quad \Rightarrow \quad \mathrm{I}(\widetilde{Z}_s; Z_s \mid C) = 0. \tag{37}$$

Then, by applying the chain rule of mutual information, we obtain

$$\mathrm{I}(\widetilde{Z}_s; Z_s, C) = \mathrm{I}(\widetilde{Z}_s; C), \tag{38}$$

and

$$\mathrm{I}(\widetilde{Z}_s; Z_s, C) = \mathrm{I}(\widetilde{Z}_s; Z_s) + \mathrm{I}(\widetilde{Z}_s; C \mid Z_s) \ge \mathrm{I}(\widetilde{Z}_s; Z_s). \tag{39}$$

Combining Eq. (38) and Eq. (39), we obtain the inequality as

$$\mathrm{I}(\widetilde{Z}_s; C) \ge \mathrm{I}(\widetilde{Z}_s; Z_s). \tag{40}$$

Meanwhile, by definition of mutual information, we have

$$\mathrm{I}(\widetilde{Z}_s; Z_s) = \mathrm{H}(Z_s) - \mathrm{H}(Z_s \mid \widetilde{Z}_s), \tag{41}$$

where

$$\mathrm{H}(Z_s \mid \widetilde{Z}_s) = \mathrm{H}(Z_s - \widetilde{Z}_s \mid \widetilde{Z}_s) \le \mathrm{H}(Z_s - \widetilde{Z}_s). \tag{42}$$

Since $\mathbb{E}||\widetilde{Z}_s - Z_s||^2 < \infty$, according to the maximum entropy principle, $\mathrm{H}(Z_s - \widetilde{Z}_s)$ reaches its maximum when $Z_s - \widetilde{Z}_s \sim \mathcal{N}(\mu, \Sigma)$. Therefore, we have

$$\mathrm{H}(Z_s - \widetilde{Z}_s) \le \mathrm{H}_{max} = \frac{1}{2}\log\det\big(2\pi e\Sigma\big) = \frac{Td}{2}\log(2\pi e) + \frac{1}{2}\log\det(\Sigma), \tag{43}$$

where Σ is the covariance matrix of $Z_s - \widetilde{Z}_s$ and $\det(\cdot)$ represents the determinant of a matrix. According to Hadamard's inequality, the determinant of Σ satisfies

$$\det(\Sigma) \le \Pi_{i=1}^{Td}\Sigma_{ii}, \tag{44}$$

thereby we have

$$\log\det(\Sigma) \le \sum_{i=1}^{Td}\log\Sigma_{ii}. \tag{45}$$

Since $\log(\cdot)$ is a concave function, according to Jensen's inequality, we have

$$\sum_{i=1}^{Td}\log\Sigma_{ii} \le Td\log\left(\frac{1}{Td}\sum_{i=1}^{Td}\Sigma_{ii}\right) = Td\log\left(\frac{\mathrm{tr}(\Sigma)}{Td}\right), \tag{46}$$

where $\mathrm{tr}(\cdot)$ represents the trace of a matrix. Note that we also have

$$\mathbb{E}||\widetilde{Z} - Z||^2 = \mathrm{tr}(\Sigma) + ||\mathbb{E}(\widetilde{Z} - Z)||^2, \tag{47}$$

which follows that

$$\mathrm{tr}(\Sigma) \le \mathbb{E}||\widetilde{Z} - Z||^2. \tag{48}$$

Substituting Eqs. (45)-(46) and Eq. (48) back to Eq. (43), we obtain

$$\mathrm{H}(Z_s - \widetilde{Z}_s) \le \frac{Td}{2}\log\left(2\pi e\frac{\mathbb{E}||\widetilde{Z} - Z||^2}{Td}\right). \tag{49}$$

Finally, substituting Eq. (49) and Eq. (41) to Eq. (40), we have the lower bound of the mutual information between $\widetilde{Z}_s$ and C as

$$\mathrm{I}(\widetilde{Z}_s; C) \geq \mathrm{H}(Z_s) - \frac{Td}{2} \log \left(2\pi e \frac{\mathbb{E}||\widetilde{Z}_s - Z_s||^2}{Td} \right) \tag{50}$$

Similarly, we can derive the mutual information lower bound for the VLA view as

$$\mathrm{I}(\widetilde{Z}_v; C) \geq \mathrm{H}(Z_v) - \frac{Td}{2} \log \left(2\pi e \frac{\mathbb{E}||\widetilde{Z}_v - Z_v||^2}{Td} \right) \tag{51}$$

A.4 Proof of Proposition 2

Proof. To optimize Eq. (13) using SGD, the parameters $\theta^{(k)}$ is first updated following the maximization objective $\max D_v$, formulated as

$$\theta^{(k+\frac{1}{2})} = \theta^k + \eta \nabla_\theta D_v(\theta^k). \tag{52}$$

Next, the parameters $\theta^{k+\frac{1}{2}}$ are further updated in the adversarial direction by minimizing D_s, following

$$\theta^{k+1} = \theta^{k+\frac{1}{2}} - \eta \nabla_\theta D_s(\theta^{k+\frac{1}{2}}). \tag{53}$$

Following the SGD paradigm, we apply first-order Taylor expansion on the gradient in Eq. (53) to obtain

$$\begin{aligned} \nabla_\theta D_s(\theta^{k+\frac{1}{2}}) &= \nabla_\theta D_s(\theta^k + \eta \nabla_\theta D_v(\theta^k)) \\ &= \nabla_\theta D_s(\theta^k) + \eta \nabla_\theta^2 D_s(\theta^k) \nabla_\theta D_v(\theta^k) + O(\eta^2), \end{aligned} \tag{54}$$

where $O(\eta^2)$ is the two-order infinitesimal that is usually ignored in first-order optimization like SGD. Therefore, by combining Eqs. (52)-(54) and omit the shared notation θ^k for simplification, we have

$$\theta^{k+1} = \theta^k - \eta(\nabla_\theta D_s - \nabla_\theta D_v + \eta \nabla_\theta^2 D_s D_v). \tag{55}$$

Let

$$g_s \triangleq \nabla_\theta D_s, \quad g_v \triangleq -\nabla_\theta D_v, \tag{56}$$

and

$$\Delta_{sv} \triangleq -\eta \nabla_\theta^2 D_s \nabla_\theta D_v = -\eta \nabla_\theta^2 D_s g_v \tag{57}$$

Then Proposition 2 follows. □